Unmasking Fraud: A Glass-Box Machine Learning Architecture for Bank Account Security

Technical Capstone Defense | Post Graduate Diploma in Data Science

Asian Institute of Management (AIM)

The 1M Record Challenge: Finding the Signal in Extreme Imbalance

High accuracy is trivial; high recall is computationally and mathematically complex.

1M

Total Bank Transactions Analyzed

Engineered Variables

29 custom features built across chronological velocity windows, structural address histories, and high-risk credit markers.

Base Fraud Rate

1.1%

Data Integrity Pipeline: Unmasking Hidden Placeholders

Detection (The Mask)

System `df.info()` reported zero missing values. Anomaly analysis exposed hidden `-1` values masquerading as valid historical entries in chronological features.

Identified -1 Placeholders in Address & Tenure metrics

Intervention (Transformation)

Targeted programmatic replacement of hidden invalid markers with explicit NaN representations to prepare features for robust imputation.

```
df.replace({
    'prev_address_months_count': -1,
    'current_address_months_count': -1,
    'bank_months_count': -1
}, np.nan, inplace=True)
```

Restoration (Imputation)

Median imputation applied to securely restore data structural integrity, preserving core distributional shapes without information loss.

Established Clean Baseline for LightGBM

Feature Engineering: Synthesizing Behavioral Signals

Raw Timestamps & Amounts → **Rolling Window Synthesis** → `velocity_6h, velocity_24h, velocity_4w`

Captures the temporal urgency and transaction frequency of malicious account exploitation over distinct chronological windows.

Income + Credit Risk Score → **Socio-Economic Interaction** → `income_x_credit_score`

Combines macroeconomic indicators to mathematically isolate high-risk financial profiles previously masked when viewed independently.

Proving Non-Linearity: Why Gradient Boosting is a Mathematical Necessity

PCA Analysis: Linear Failure

First two principal components capture ~100% variance, yet completely fail to linearly separate fraud from legitimate accounts.

Linear separation impossible in principal component space.

UMAP Projection: Manifold Discovery

UMAP reveals intricate, highly granular non-linear manifold structures and localized behavioral clusters.

Complex non-linear decision boundaries required.

The Modeling Arena: Baseline to Ensemble Architecture

Algorithm	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.7117	0.0261	0.6924	0.0503	0.7707
Decision Tree	0.9785	0.0452	0.0471	0.0462	0.5180
Random Forest	0.9890	0.5000	0.0003	0.0006	0.7301
Linear SVC	0.9889	0.1111	0.0003	0.0006	0.7707
XGBoost	0.8422	0.0424	0.6159	0.0793	0.8230
LightGBM (Tuned)	**0.7792**	**0.0355**	**0.7262**	**0.0676**	**0.8325**

Verdict: Tuned LightGBM selected as champion model. Balances extreme class imbalance effectively, yielding dominant AUC of 0.8325.

Hyperparameter Optimization: Navigating the Solution Space with Optuna

Optimization Trajectory

Executed 50 automated algorithmic trials to systematically maximize the AUC metric on the validation set, utilizing stochastic tree-building parameters.

0.8325

Maximized Validation AUC

The Champion Configuration

`n_estimators: 1553`
Extensive tree building phase

`learning_rate: 0.0116`
Slow, precise gradient descent

`num_leaves: 20 & max_depth: 12`
Controlled complexity to prevent overfitting

The Evaluation Trade-Off: Maximizing Recall vs. False Positives

Untuned Baseline Model

253,926 True Negatives	42,765 False Positives
1,360 False Negatives (Missed)	1,949 True Positives (Caught)

Optuna-Tuned LightGBM

231,348 True Negatives	65,343 False Positives
906 False Negatives (Missed)	2,403 True Positives (Caught)

Business Reality: Tuning successfully minimized dangerous False Negatives from 1,360 to 906, significantly expanding fraud interception capacity.